

SYAFRIZAL HIDAYAT

Bandung, Indonesia • +62 853-6383-4049 • hidayatsyafrizal@gmail.com • [linkedin.com/in/syafrizal-hidayat](https://www.linkedin.com/in/syafrizal-hidayat)

PROFESSIONAL SUMMARY

Cum Laude graduate of Institut Teknologi Bandung's Oceanography program with end-to-end capability across three pillars: marine field operations, geospatial software development, and oceanographic data analysis. Hands-on experience executing offshore hydrographic expeditions (SBES, GNSS RTK, CTD, WQM, Current Meter, Nivo Leveling) at Pramuka Islands, submarine morphology data analysis at Batu Tara, and geophysical field campaigns at the Dieng Volcanic Complex paralleled by production WEBGIS platform delivery (Python, GeoDjango, Leaflet.js) for BRIN and KKP. Holder of 2 registered Intellectual Property Rights (HAKI), with proven leadership and communication credentials directing protocol for 15,000+ attendee university events, serving as Master of Ceremonies for major programs, founding a 130-member humanitarian community, and sustaining a 3-year volunteer teaching commitment for underprivileged students. Open to roles bridging marine operations, geospatial technology, applied research, and team coordination.

EDUCATION

Bachelor of Science in Oceanography

Aug 2022 — Aug 2026

Institut Teknologi Bandung (ITB)

GPA: 3.74 / 4.00 (Cum Laude Candidate) • 4× Dean's List Awardee

- Dual Scholarship Recipient: National KIP Kuliah (KIPK) Grant & PT Timah Tbk Scholarship.
- Final-Year Thesis (KKP Collaboration): Developed an interactive WEBGIS geoportal for Central Kalimantan, integrating real-time metocean data (CMEMS, ECMWF) with coastal zonation regulations using GeoDjango and Leaflet.js.
- Active across multiple semesters as Teaching & Practicum Assistant (Numerical Methods, Marine Data Statistics).

SELECTED PROFESSIONAL EXPERIENCE

Geospatial Web Developer & Field Researcher

Oct 2025 — Present

National Research and Innovation Agency (BRIN) — DiVolca.Net

- Engineered the "DiVolca.Net" WEBGIS prototype for physical volcanology, supporting national disaster mitigation and early warning systems.
- Executed geophysical field campaigns at the Dieng Volcanic Complex; processed UAV photogrammetry into 3D spatial models and DEMs.
- Contributed to 2D/3D bathymetric and topographic visualization pipelines supporting scientific manuscript publication.

Marine Geophysics Intern

Mar — Apr 2026

PT. Prihaditama Geosciences and Oceanography Consulting

- Processed offshore Sub-Bottom Profiler (SBP) and onshore Ground Penetrating Radar (GPR) data using ReflexW; detected underground pipeline positioning for client engineering planning.
- Drafted 2D utility alignments and subsurface boundaries in AutoCAD as final deliverables.

Research Assistant & WEBGIS Developer

Sep 2024 — Dec 2025

Central Kalimantan Geoportal Project (ITB × KKP)

- Built an interactive WEBGIS geoportal in collaboration with the Ministry of Marine and Fisheries (KKP), using GeoDjango (backend) and Leaflet.js (frontend) for regional coastal spatial planning.
- Integrated real-time metocean data (currents, waves, winds) from CMEMS and ECMWF APIs into the spatial database, supporting government stakeholder access.

WEBGIS Developer & IoT Instrumentation Intern

Jun 2024 — Aug 2025

PT. Brajakara Teknologi Media

- Implemented dynamic mapping features including animated vector fields for ocean currents and temporal Sea Surface Temperature (SST) heatmaps using Python.
- Assembled Arduino-based microcontroller systems integrated with temperature sensors and RTCs for environmental monitoring applications.

- Processed raw CTD oceanographic data and built frontend interfaces with complete CRUD operations.

SELECTED PROJECTS

VARUNA WATCH — Coastal Environmental Sensor (HAKI Awarded)

Mar — Sep 2025

Lead Instrument Engineer — Selayar Islands

- Engineered a real-time sea level sensor for tsunami mitigation and climate adaptation; secured Intellectual Property Rights (HAKI, DJKI, Sep 2025).
- Co-authored "Menjelajahi Laut dan Mengenal Bencananya" educational book on oceanography and coastal disaster mitigation (ISBN: 978-623-127-880-7, May 2026).

Submarine Morphological Analysis of Batu Tara Volcano

Jan — Apr 2026

Research Assistant — Flores Sea

- Processed Multibeam Echosounder (MBES) data and generated 20-meter resolution DEM using Python (Rasterio, NumPy, Cartopy, Matplotlib).
- Conducted slope and terrain analysis for scientific manuscript publication on volcanic morphology.

Oceanographic Field Survey Expedition — Pramuka Islands

May 2025

Hydrographic Surveyor

- 10-day intensive offshore and coastal expedition operating SBES, CTD, Current Meter, WQM, GNSS RTK, and handheld GPS for high-precision coastal mapping and data acquisition.

LPDP EQUITY Community Development Program (HAKI Output)

Oct 2025 — May 2026

Technical Assistant — Cirebon

- Co-engineered a solar-powered seaweed drying oven for Muara Village, Cirebon — outcome registered as Intellectual Property Rights (HAKI, DJKI, Apr 2026).
- Conducted water quality field surveys in seaweed aquaculture ponds using WQM Horiba and GPS.

TEACHING & MENTORSHIP

Teaching Assistant — Numerical Methods & Marine Data Statistics

Aug 2024 — Jul 2025

Institut Teknologi Bandung (ITB)

- Facilitated 14 combined practicum sessions across two oceanography courses, covering ANOVA, regression, forecasting, ODE/PDE solvers, and numerical integration using Python, Matlab, and Fortran.

Physics Olympiad Coach & Private Tutor

Sep 2025 — Present

ALC Indonesia & Bimbel Jinje Smart Academy

- Coached Olympiad-level Physics for SMA/SMP students across 4 partner schools in West Java, Central Kalimantan, and East Java; ongoing private Math & Physics tutoring (online).

Volunteer UTBK Instructor

2023 — 2026

Sakola Kembara — Cirebon, Cililin, Bandung

- 3-year pro-bono coaching for underprivileged high school students across 3 cities, demonstrating sustained social commitment alongside academic and professional responsibilities.

LEADERSHIP & COMMUNITY ENGAGEMENT

Founder & Community Lead

Jun 2025 — Present

Hidayahsubuh.id — Humanitarian Community

- Founded and lead a 130-member humanitarian community under the tagline "Sedekah Subuh 1000 Rupiah"; executed 20+ social programs (Qurban, orphanage outreach, sembako distribution).

Core Protocol Officer

Jul 2024 — Present

Protokol ITB Team

- Coordinated protocol for university events serving 15,000+ attendees per event (Graduation Ceremonies, Studium Generale); served as primary liaison for VIP guests and university leadership.

Operational Coordinator

Dec 2023 — Feb 2025

Character Development Training (CDT) ITB 2024

- Commanded 50 field committee members for a 5,000-student orientation program; designed event rundowns and served as main MC.

TECHNICAL SKILLS & COMPETENCIES

Field Survey Operations: Single-Beam Echosounder (SBES), GNSS RTK, CTD, WQM Horiba, Current Meter, Nivo Leveling, Handheld GPS, UAV/Drone Photogrammetry

Geophysical Data Processing: Multibeam Echosounder (MBES), Sub-Bottom Profiler (SBP), Ground Penetrating Radar (GPR), Submarine Morphology Analysis

Geospatial Software & WEBGIS: Python, GeoDjango, Django, Leaflet.js, JavaScript, HTML/CSS, Spatial Database (PostGIS), CRUD Operations

Data Analysis & Computing: Rasterio, NumPy, Cartopy, Matplotlib, Pandas, Matlab, Fortran, Excel; ANOVA, Regression, Forecasting, Numerical Methods

Hardware & IoT: Arduino Microcontrollers, Sensor Integration, Solar-Powered Systems, Proteus, Tinkercad

Geospatial Tools: ArcGIS, QGIS, Surfer, Ocean Data View (ODV), ReflexW, AutoCAD, HYPACK

Soft Skills: Project & Event Management, Team Leadership, Public Speaking, Stakeholder Coordination, Master of Ceremonies

Languages: Indonesian (Native Proficiency), English (Intermediate)

HONORS & ADDITIONAL INFORMATION

- Intellectual Property Rights (HAKI) — VARUNA WATCH coastal environmental sensor (DJKI, Sep 2025).
- Intellectual Property Rights (HAKI) — Co-creator, Solar-Powered Seaweed Drying Oven System for Muara Village, Cirebon (DJKI, Apr 2026).
- Co-author — "Menjelajahi Laut dan Mengenal Bencananya" educational book (ISBN: 978-623-127-880-7, May 2026).
- Cum Laude Candidate, GPA 3.74/4.00 — 4× Dean's List Awardee at Institut Teknologi Bandung.
- Dual Scholarship Recipient — National KIP Kuliah (KIPK) & PT Timah Tbk Scholarship.
- Official ITB Delegate — Startup Ecosystem Program with Tarumanagara × NUS Singapore (Jan 2025); National KIPK Forum at Ministry of Education and Culture (May 2024).
- Certified Tahsin — Fashahatul Lisan Method ('Maqbul' predicate), Pusat Studi Al-Quran (May 2024).